

Nonstationarity Manufactures Heavy Tails: Scale-Normalized Sequential Learning

Anonymous Authors¹

Abstract

We study online prediction on streams whose loss-gradient *scale* is strongly nonstationary, using a large financial dataset as a testbed. The gradient-norm scale drifts several-fold across the record with overnight jumps, and this drift alone makes the *pooled* gradient look heavy-tailed (tail index $\hat{\alpha} \approx 2.4$); the *causally scale-normalized* gradient—the quantity a learner should actually act on—is markedly lighter ($\hat{\alpha} \approx 2.9$ – 3.7 , estimator-dependent), so the difficulty is dominated by nonstationarity rather than by an intrinsically heavy tail. In this regime plain online gradient descent (OGD) has no usable learning rate—small rates underfit while any rate large enough to learn diverges when the scale spikes—and clipping to a *scale-dependent* threshold is fragile because the clipped step still inherits the drift. We propose *Scale-Normalized OMD* (SN-OMD), which divides the gradient by a tracked robust scale and caps it at a *scale-free* constant M ; the per-step move is then bounded by M rather than by a regime-dependent quantity, so the iterates provably remain bounded for any learning rate or realization (a stability, not convergence, guarantee). A dynamic-regret analysis—attaining the per-regime rate $T^{1/p}$ under a finite, possibly time-varying p -th moment ($p \in (1, 2]$)—is deferred to the appendix, since on real data the scale drift enters it as a genuine $\sqrt{T}$ factor; we lead instead with the empirically-testable mechanism. Empirically, scale-invariant updates (SN-OMD and the classical normalized-GD alike) stay bounded and accurate across a wide learning-rate range where OGD and scale-dependent clipping diverge; SN-OMD adds, over normalized-GD, a regret guarantee via its *predictable* scale and—

through the cap—non-divergence at large steps and an accuracy edge when the tail survives normalization.

1. Introduction

Financial markets are a canonical source of heavy-tailed, distribution-free data: returns and microstructure signals exhibit power-law tails, occasional data errors act as adversarial contamination, and assumptions such as sub-Gaussianity are untenable. Sequential prediction on such streams therefore calls for *distribution-free* guarantees that hold under weak moment assumptions only. A rich line of work in robust statistics shows that the empirical mean is a poor estimator under heavy tails, whereas estimators such as Catoni’s M-estimator, median-of-means, and the trimmed mean achieve sub-Gaussian deviations assuming only a finite variance (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); in optimization, gradient clipping is the corresponding algorithmic primitive for heavy-tailed gradient noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023).

This paper concerns a facet these lines mostly abstract away: on real streams the gradient *scale* is strongly nonstationary, and this—more than an intrinsically heavy tail—is what breaks existing methods. Measuring the gradient of an online linear predictor on the Jane Street market dataset, we find (i) the gradient *scale* drifts roughly $6\times$ across two months with overnight jumps; (ii) this drift alone inflates the *pooled* gradient-norm tail index to $\hat{\alpha} \approx 2.4$ —a scale mixture of lighter-tailed increments looks heavy-tailed—whereas the *causally scale-normalized* gradient $\|g_t\|/s_t$, which is what a learner actually acts on, is much lighter, $\hat{\alpha} \approx 2.9$ – 3.7 ; and (iii) the *gradient* tail is nonetheless heavier than either its feature or residual factor alone, an *interaction* effect from their co-movement. The operative difficulty is thus nonstationarity of scale, with only a mild residual heavy tail. This regime—a drifting scale against a moving comparator—is exactly where existing distribution-free online learning results, which are either in expectation, on bounded domains, or against a static comparator (Zhang et al., 2026; Zhang & Cutkosky, 2022), do not reach.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

The consequences are sharp. Plain OGD has *no* usable learning rate on such a stream: a small rate underfits, while any rate large enough to learn diverges the first time the gradient scale spikes in a turbulent regime. The textbook fix—clip the gradient at a threshold τ_t tracking the local scale—does not help under a drifting scale, because the clipped step is $\propto \min(\|g_t\|, \tau_t)$ and so still scales *with* the regime: a turbulent window produces large steps and diverges. We show this failure both empirically and as a structural property of any scale-dependent threshold.

Our approach. The fix is to decouple the step from the tail scale. *Scale-Normalized OMD* (SN-OMD) divides the gradient by a predictable robust scale s_t and caps the result at a *scale-free* constant M ,

$$\hat{g}_t = \text{clip}(g_t/s_t, M), \quad w_{t+1} = \Pi_{\mathcal{W}}\left(w_t - \frac{\eta}{\sqrt{t}} \hat{g}_t\right).$$

The single factor $1/s_t$ replaces clipping’s *scale-dependent* step bound $M s_t$ with a *scale-free* bound M . Because $\|\hat{g}_t\| \leq M$ deterministically, the iterates remain bounded for *any* learning rate or realization—an unconditional *stability* guarantee (not, on its own, convergence) that needs no moment assumption.

Contributions.

- **Nonstationarity manufactures heavy tails (Section 4).** On a large market dataset the gradient scale drifts $\sim 6\times$; this drift alone makes the *pooled* gradient tail heavy ($\hat{\alpha} \approx 2.4$), while the scale-normalized gradient is much lighter ($\hat{\alpha} \approx 2.9\text{--}3.7$)—an effect we confirm with an *exogenous* scale, and which recurs, in weaker form, on image-model training gradients. This is the paper’s central finding.
- **Algorithm, stability, and a cap frontier (Sections 3 and 4.1).** We propose SN-OMD, prove its iterates are unconditionally bounded (Theorem 3.1), and show it is a one-parameter family in the cap M interpolating normalized-GD ($M \rightarrow 0$) and uncapped scale-adaptive OGD ($M \rightarrow \infty$); a two-directional test predicts *when* the finite cap matters—non-divergence at large steps, and an accuracy optimum only when the normalized tail is genuinely heavy.
- **A protocol-careful study, and deferred theory (Section 4).** Under causal standardization SN-OMD stays bounded and most accurate where OGD and scale-dependent clipping diverge; it is invariant to per-row vs. per-timestep updates (no contemporaneous leakage) and its advantage vanishes on a genuinely stationary control. A per-regime dynamic-regret bound (Theorem A.2) is deferred to the appendix, honest about a $\sqrt{T}$ drift factor; a tested implementation is released.

2. Problem Setup

Protocol. Learning proceeds in rounds $t = 1, \dots, T$. At round t the learner plays $w_t \in \mathcal{W} \subseteq \mathbb{R}^d$ (a convex set of diameter D ; the unbounded case is discussed in Section 3), observes a convex loss ℓ_t (for us the sample-weighted squared error $\ell_t(w) = \omega_t(\langle w, x_t \rangle - y_t)^2$), and receives a stochastic subgradient g_t with $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = \bar{g}_t \in \partial \ell_t(w_t)$, where $(\mathcal{F}_t)$ is the natural filtration and $\varepsilon_t = g_t - \bar{g}_t$ is the noise.

Drifting heavy-tailed gradients. We assume only a *finite, time-varying* low-order moment of the subgradient itself—the quantity we measure and the algorithm acts on:

Assumption 2.1 (Drifting p -th moment). For each t , $\mathbb{E}[\|g_t\|^{p_t} \mid \mathcal{F}_{t-1}] \leq \sigma_t^{p_t}$ for some $p_t \in (1, 2]$, where both the scale σ_t and the tail index p_t may drift with t , and $p_t < 2$ (infinite variance) is permitted. Write $p := \inf_t p_t$. (This bounds $\bar{g}_t$ and the noise $\varepsilon_t = g_t - \bar{g}_t$ up to constants, and is exactly the tail we estimate in Section 4.)

No sub-Gaussianity, boundedness, or stationarity is assumed. *Distribution-free* means the learner knows none of σ_t, p_t, D , or the comparator norm. The regret analysis does additionally require a *predictable* scale tracker (Assumption A.1); we do not treat this as free, and discharge it under a mild drift model (Proposition B.5).

Dynamic regret. Performance is measured against a moving comparator $u_{1:T}$ with path length $P_T = \sum_t \|u_t - u_{t-1}\|$,

$$R_T(u_{1:T}) = \sum_{t=1}^T [\ell_t(w_t) - \ell_t(u_t)] \leq \sum_{t=1}^T \langle \bar{g}_t, w_t - u_t \rangle, \quad (1)$$

the notion appropriate for regime-switching streams; static regret is the special case $u_t \equiv u$. We report the sample-weighted zero-mean $R^2 = 1 - \sum_i \omega_i (y_i - \hat{y}_i)^2 / \sum_i \omega_i y_i^2$ on data.

2.1. A budget heuristic for the dynamic wall

The achievable rate degrades smoothly with the tail. The core primitive is robust mean estimation: under a finite p -th moment the best estimator of a mean from n samples has error $\Theta(\sigma n^{-(1-1/p)})$ (Devroye et al., 2016; Lugosi & Mendelson, 2019), which interpolates the sub-Gaussian rate ($p = 2$) and no concentration ($p \rightarrow 1$). Aggregated online this yields, and is matched by, a regret of order $T^{1/p}$ (Zhang et al., 2026):

$$R_T \asymp \sigma \cdot T^{1/p}, \quad T^{1/2} (p=2) \longrightarrow T (p \rightarrow 1). \quad (2)$$

Two refinements matter under nonstationarity. Partitioning $[T]$ into regimes r of length L_r , index p_r , and scale σ_r , and applying the bound within each, *static* regret (one comparator) is governed by the worst regime, while *dynamic* regret

(a comparator per regime) has the per-regime costs *add*:

$$R_T^{\text{stat}} \gtrsim \max_r \sigma_r L_r^{1/p_r}, \quad R_T^{\text{dyn}} \gtrsim \sum_r \sigma_r L_r^{1/p_r}. \quad (3)$$

Because $L^{1/p}$ is convex, heavy-tail cost cannot be averaged across regimes: a few short heavy regimes ($p_r < 2$) dominate the sum. Two caveats keep (3) honest. First, it is a per-regime *budget*, not a proved regret lower bound at a fixed path length: it charges each regime its worst-case cost against a *fresh* comparator, so it lower-bounds dynamic regret only against an *unconstrained* comparator sequence—which is trivially $\Omega(T)$ in any case. The interesting object is regret at a given path length P_T , which (3) does not constrain. Second, it separates drift from the static case only qualitatively: prior heavy-tailed results fix a *single* p , a stationary scale, and a *static* comparator, and recover the $T^{1/p}$ rate (2) (Zhang et al., 2026; Zhang & Cutkosky, 2022; Bubeck et al., 2013), whereas here the per-regime costs *add* across a drifting exponent, and the static budget (a max, not a sum) does not see this. Third—and this is why we keep (3) *qualitative*—the sum is dominated by regimes with $p_r < 2$, which we measure only on *pooled* gradients; after scale normalization the tail the learner actually sees has $p \approx 2$ (Section 4), and the super- $\sqrt{T}$ growth of the pooled budget largely evaporates. We therefore read (3) as a *motivating heuristic*—the interaction of tail drift with a moving comparator is what makes the dynamic problem hard—and not as a measured wall; a proper P_T -constrained Ω is left open. What SN-OMD delivers is to attain the per-regime worst-tail rate $T^{1/p}$ rather than diverge.

3. Scale-Normalized Online Mirror Descent

Algorithm. SN-OMD maintains a *predictable* robust estimate $s_t > 0$ of the gradient-norm scale (an $\mathcal{F}_{t-1}$ -measurable function of past gradient norms) and takes a scale-invariant, constant-capped step (Algorithm 1). Writing $v_t = g_t/s_t$ and $\text{clip}(v, M) = v \cdot \min\{1, M/\|v\|\}$, the update is $w_{t+1} = \Pi_{\mathcal{W}}(w_t - \frac{\eta}{\sqrt{t}} \text{clip}(v_t, M))$. The scale s_t can be any predictable robust tracker; we use a winsorized exponential moving average of $\|g_t\|$ by default, and analyze a two-timescale “peak-hold” variant below.

Why normalize, not clip. Clipping to a scale-dependent threshold $\tau_t = c s_t$ gives a step $\propto \min\{\|g_t\|, c s_t\}$, which scales *with* the drifting s_t : a turbulent regime yields large steps and divergence. Dividing by s_t first gives a step $\propto \min\{\|g_t\|/s_t, M\}$, bounded by the *scale-free* constant M . The two differ by the factor $1/s_t$, and this single factor is what removes the scale-drift transmission.

Theorem 3.1 (Stability, unconditional). *For any inputs, $\|\hat{g}_t\| \leq M$ deterministically. Hence on a bounded domain $\|w_t\| \leq D$ for all t , and on $\mathbb{R}^d$ (step $\eta/\sqrt{t}$) $\|w_t\| \leq \|w_1\| +$*

Algorithm 1 Scale-Normalized OMD (SN-OMD)

Input: cap M , step η , predictable scale tracker $\mathcal{S}$; $w_1 = 0$.

for $t = 1$ **to** T **do**

 play w_t ; observe ℓ_t and stochastic subgradient g_t
 $s_t \leftarrow \mathcal{S}.\text{scale}$ (predictable, uses g_t for $t' < t$)
 $\hat{g}_t \leftarrow \text{clip}(g_t/s_t, M)$
 $w_{t+1} \leftarrow \Pi_{\mathcal{W}}(w_t - \frac{\eta}{\sqrt{t}} \hat{g}_t)$
 $\mathcal{S}.\text{update}(\|g_t\|)$

end for

$2M\eta\sqrt{t}$. Hence the iterates remain uniformly bounded (on $\mathcal{W}$) or grow at most as $O(\sqrt{t})$ (on $\mathbb{R}^d$) for any learning rate, scale drift, or realization—a stability guarantee, not by itself convergence, that uses no moment assumption.

This is the crucial contrast with scale-dependent clipping, whose per-step move $\propto s_t$ is bounded only by the drifting scale.

Dynamic regret (deferred to the appendix). Beyond stability, SN-OMD admits a high-probability *dynamic*-regret analysis (Section A): under a scale-tracking condition—a predictable s_t bracketing the local scale (Assumption A.1)—it attains the per-regime rate $T^{1/p}$ under only a finite p -th moment. We keep it in the appendix, not the headline, for two honest reasons. The bound is stated through the tracker’s upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$, which we *measure* to grow linearly with T on real data (Section 4.3); so the scale drift is a genuine $\sqrt{T}$ factor and the $T^{1/p}$ rate holds per regime, not globally. And the scale-tracking condition, though one-sided and empirically dischargeable (Proposition B.5), remains an assumption. What the body develops instead is the empirically-testable content—why normalizing beats clipping, and where the cap earns its keep.

4. Experiments

Setup and protocol. We use the Jane Street Real-Time Market Data Forecasting dataset (Jane Street, 2024): 47,127,338 rows over 1,699 trading days, 79 features, 39 symbols; target `responder_6`; metric the sample-weighted zero-mean R^2 . We evaluate *progressive online* prediction on the historical record: each round the learner plays w_t , then observes (x_t, y_t) and updates—the protocol our theory concerns. Features are standardized *causally* (running past-only statistics; a full-sample standardizer inflates R^2 by look-ahead and is not used). We report two round granularities: *per-row* and *per-timestep* (predict the whole (date, time) cross-section of ~ 9 symbols, then one aggregated update), the latter carrying no contemporaneous cross-sectional information. **This is not the competition’s**

protocol (batched timesteps served with day-lagged responders, scored on a held-out future window), so our R^2 is not comparable to competition leaderboard scores in either direction; we report it only for cross-method comparison and the stability contrast. Tail measurements are learner-independent (gradients at the fixed best linear predictor w^*).

The scale is nonstationary; the pooled tail is mostly a scale mixture. At the fixed optimum w^* , the *pooled* gradient-norm tail index is $\hat{\alpha} \approx 2.43$, and it is an *interaction* effect: the residual and feature norms each have $\hat{\alpha} \approx 4$, but their product—the gradient—has a heavier tail because extreme features and residuals co-occur (Figure 1a). Most of that pooled heaviness, however, is manufactured by nonstationarity. The daily gradient scale drifts $\approx 6\times$ with overnight jumps (Figure 1b), and once we divide by a robust scale s_t —the quantity SN-OMD actually acts on—the tail lightens markedly, to $\hat{\alpha} \approx 3.7$ (causal winsorized-EMA s_t), ≈ 3.2 (non-causal median), or ≈ 2.9 (trailing median), i.e. toward finite higher moments. This is not merely a self-normalization artifact: dividing by an *exogenous* scale that never sees the gradient magnitude (an EMA of the feature norm $\|x_t\|$) still lightens the tail to $\hat{\alpha} \approx 2.9$. So the operative difficulty is the scale drift, leaving only a mild residual heavy tail. (Per-day $\hat{\alpha}$ dips below 2 on a minority of days, but single-day Hill estimates are noisy and we do not rely on them.) We note that the per-regime budget of Section 2.1 is dominated by exactly these pooled- $\alpha < 2$ days; since those regimes are themselves a pooled-tail artifact—they largely disappear once the gradient is normalized (the tail the learner sees has $p \approx 2$)—we treat that budget only as qualitative motivation for why a moving comparator under drift is hard, not as a measured quantity.

A second domain, and its limit. On the per-step gradient norm of a small CNN trained on MNIST (2,814 steps), the pooled tail is heavy ($\hat{\alpha} \approx 2.6$) and normalizing lightens it ($\hat{\alpha} \approx 6$; single-window Hill estimates are unstable here, 5–9 across thresholds). But this is *schedule-driven*: within a single constant-learning-rate window (nearly flat scale) the tail is *already* light ($\hat{\alpha} \approx 6$ –9), so the pooled heaviness comes from the scale decay and the learning-rate drops, not from within-phase drift. The drift-manufactures-tails effect thus recurs on training gradients, but only in this weaker, transient form; the genuine within-regime version—confirmed by an *exogenous* scale and a feature–residual interaction—we establish only on the market data.

Scale-invariant updates dominate scale-dependent ones. Streaming one learner through the multi-regime window and sweeping the learning rate (Figure 3, Table 1), the methods split cleanly by whether their step responds to the gradient *scale*. *Scale-dependent* methods—plain OGD and the robust clippers (AdaptiveClip, RobustOMD), whose

threshold $\tau_t = c s_t$ rescales with the regime—are fragile: they stay bounded only up to $\text{lr} \approx 10^{-2}$ and reach best $R^2 \approx 0.01$ –0.14, because any larger rate diverges when the scale spikes (Figure 2). *Scale-invariant* methods—normalized-GD and SN-OMD (bounded to $\text{lr} = 2$) and uncapped scale-adaptive OGD (to $\text{lr} = 1$)—reach $R^2 \approx 0.08$ –0.32 across the two protocols. Per-timestep batching lifts OGD from 0.02 to 0.14 (averaging the ~ 9 correlated symbols per timestep) but leaves it fragile above $\text{lr} \approx 10^{-2}$; SN-OMD is essentially unchanged ($0.285 \rightarrow 0.309$), so its accuracy is not a contemporaneous-information artifact. (Uncapped scale-adaptive OGD instead *halves* under batching, $0.162 \rightarrow 0.080$: aggregating ~ 9 gradients enlarges the occasional untruncated step that a finite cap would have contained—a small preview of the cap’s role.)

What the tracked scale and the cap buy. Among the scale-invariant methods, SN-OMD and normalized-GD are *comparable* on this data (0.29/0.31 per-row/ per-timestep vs. 0.20/0.32): *normalization*—dividing by the scale—is what earns the gain over the divergent baselines, and normalized-GD is a valid simple instance of it. SN-OMD’s specific contributions lie elsewhere. First, its scale is a *predictable* tracker, which is what makes the regret bound (Theorem A.2) go through; normalized-GD’s instantaneous $1/\|g_t\|$ is not predictable and discards gradient magnitude every round. Second, the cap buys *non-divergence at large steps*: uncapped scale-adaptive OGD diverges at $\text{lr} \geq 2$ (R^2 collapses to -63), whereas SN-OMD ($M=5$) stays bounded there and in fact peaks at $\text{lr} = 2$. The two are close on Jane only because the *normalized* tail is mildly heavy ($\hat{\alpha} \approx 3.7$); Section 4.1 shows that when it is genuinely heavy the cap’s accuracy edge becomes decisive.

Negative control. If the advantage is really about scale drift, it should vanish when the scale is constant. Imposing genuine stationarity in the synthetic harness ($\sigma_t \equiv 1$, $p=2$, static reachable comparator—the one stream where stationarity can be enforced), tuned SN-OMD is if anything *worse* than tuned OGD (steady excess loss 6.6 vs. 5.6×10^{-4} over 400 seeds, an 18% gap the wrong way): the advantage disappears as the thesis predicts. (A 100k-row Jane slice is *not* a valid stationarity control—it still has $\sim 30\%$ within-day drift—so SN-OMD’s lead there is expected; what it does isolate is that the clippers’ tail-rejection, unlike scale-invariance, does not even beat OGD, 0.013–0.015 vs. 0.02.)

4.1. The cap M interpolates normalized-GD and scale-adaptive OGD

SN-OMD is a one-parameter family in the cap M : when $\|g_t\|/s_t > M$, $\text{clip}(g_t/s_t, M) = M g_t/\|g_t\|$ and s_t cancels, so $M \rightarrow 0$ is *normalized-GD* and $M \rightarrow \infty$ is uncapped *scale-adaptive OGD*. Both satisfy Theorem 3.1, so

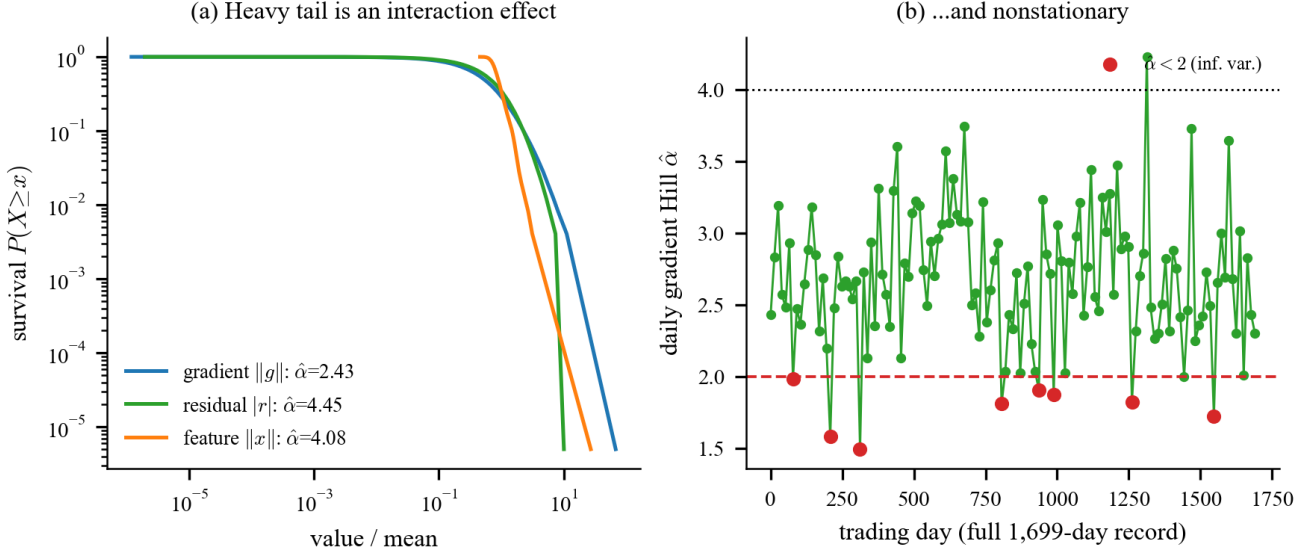


Figure 1. The scale drifts; the pooled tail is largely a scale mixture. (a) The *pooled* gradient tail (Hill index $\hat{\alpha} \approx 2.43$) is heavier than either its residual or feature factor ($\hat{\alpha} \approx 4$)—an interaction effect from co-movement. (b) The daily gradient tail index across the full 1,699-day record drifts; per-day estimates occasionally fall below 2 but single-day Hill estimates are noisy. Normalizing by the scale lightens the tail to $\hat{\alpha} \approx 2.9$ –3.7 (estimator-dependent), so the pooled heaviness is largely nonstationarity (Section 4).

Table 1. Best weighted R^2 (each method at its own tuned rate) on the continuous Jane Street stream, *causal* standardization, for both round granularities, with the largest stable learning rate. Scale-dependent methods (top) are fragile (stable only to $\text{lr} \approx 10^{-2}$); scale-invariant methods (bottom) stay bounded much higher (SN-OMD and normalized-GD to $\text{lr} = 2$, uncapped only to $\text{lr} = 1$) and are far more accurate. SN-OMD and normalized-GD are comparable here; the cap’s role is non-divergence (uncapped diverges at $\text{lr} \geq 2$). R^2 is not comparable to competition scores (different protocol, see text).

Method	R^2 per-row	R^2 per-step	Stable
<i>Scale-dependent (fragile):</i>			
OGD	0.018	0.139	$\leq 10^{-2}$
AdaptiveClip / RobustOMD	~ 0.01	—	$\leq 10^{-2}$
<i>Scale-invariant (bounded to $\text{lr} = 2$):</i>			
Normalized-GD ($M \rightarrow 0$)	0.197	0.316	≤ 2
Scale-adaptive OGD ($M \rightarrow \infty$)	0.162	0.080	≤ 1
SN-OMD ($M=5$, ours)	0.285	0.309	≤ 2

non-divergence is not what singles out SN-OMD; the point is that a finite M *preserves gradient magnitude within the bulk* (the step shrinks near a good predictor) and truncates only the tail, whereas normalized-GD discards magnitude every round. The tracked s_t makes “the bulk” regime-independent: the test $\|g_t\|/s_t \leq M$ is scale-free, which a fixed threshold cannot be under $6\times$ drift.

Sweeping M (each cap at its own tuned step, error bars over seeds) turns the cap into a *mechanism test*. Normalization already removes most of the tail ($\hat{\alpha} : 2.4 \rightarrow 3.7$), so the cap should earn its keep only when the *normalized* gradient

is still heavy—a prediction with opposite consequences in two settings. On a synthetic stream with an *imposed* heavy normalized tail ($p = 1.5$, static reachable comparator; Figure 4) an interior optimum appears: the best cap is $\approx 240\times$ better than the uncapped end (heavy-tail instability) and $\approx 10\times$ better than the normalized-GD end (magnitude-blind oscillation), both outside the seed-noise band. On Jane, where the normalized tail is only mildly heavy, there is *no* meaningful interior accuracy optimum: normalized-GD, SN-OMD, and uncapped scale-adaptive OGD are comparable (Table 1). Same mechanism, both predictions borne out. This gives the cap a precise scope: it buys *accuracy* when the tail survives normalization, and *stability unconditionally*—uncapped scale-adaptive OGD diverges at $\text{lr} \geq 2$ while SN-OMD does not (Table 1).

4.2. Synthetic streams with known (σ_t, p_t)

On synthetic streams with a controlled tail (Student- t index p ; Figure 6 in the appendix), plain OGD’s fitted cumulative-regret exponent rises $0.37 \rightarrow 0.70 \rightarrow 1.55$ as p falls $2 \rightarrow 1.3$ —super-linear (divergent) once the tail is heavy—reproducing the real-data divergence with a *known* tail index, so the tail is the mechanism, not a learning-rate artifact. SN-OMD stays sublinear, with exponents *below* $1/p$ (the stochastic stationary instance is easier than the worst case, so this is consistent with, not a tightening of, Theorem A.2); against a drifting comparator its dynamic regret grows with path length at exponent ≈ 0.43 , consistent with the $\sqrt{DP_T}$ term.

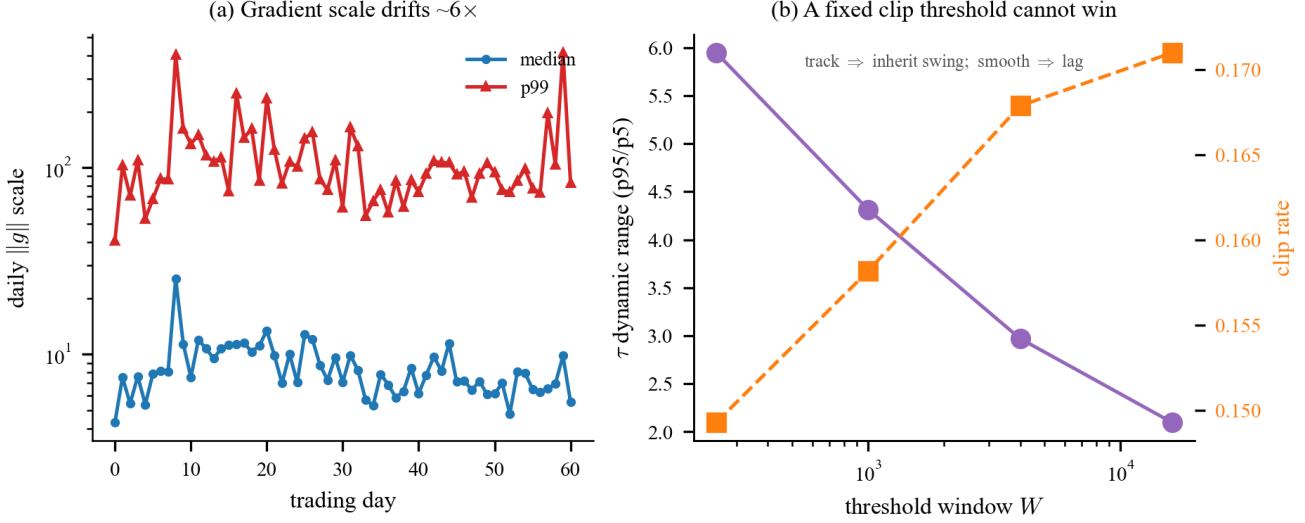


Figure 2. **Why scale-dependent clipping fails.** (a) The gradient scale drifts $\sim 6\times$ across days. (b) A fixed-window adaptive clip threshold cannot win: a short window tracks the scale (so the step inherits the swing), a long window is smooth but lags regime onsets.

4.3. Discharging the scale-tracking assumption

The regret bound (Theorem A.2) rests on Assumption A.1, that a *predictable* scale s_t brackets the local noise scale. We close this loop empirically on the real gradient-scale process (357,000 gradient norms at w^* , ground-truth scale a centered rolling median, all trackers predictable), where the true upward variation is V_σ^+ . The relevant requirement is one-sided (Section 3): hold the lower bracket $s_t \geq \frac{1}{2}\sigma_t$ everywhere, and keep the upward variation W_s (which enters Theorem A.2 as $\sqrt{W_s}$) as close to the true drift V_σ^+ as possible. A single-timescale trailing median cannot do both: a fast window tracks the scale but *churns* on heavy-tailed spikes ($W_s = 62 V_\sigma^+$), while a slow window is smooth but *lags* regime onsets and dips below the lower bracket. A two-timescale “peak-hold” envelope—react up on a short robust median, release down only geometrically—resolves the tension: it holds the stability-critical lower bracket at 100% of rounds while attaining the *smallest* upward variation, $W_s \approx 6.7 V_\sigma^+$ (Figure 5, Table 2). Its only cost is a benign 24% over-estimation of the scale (mild underfit, never divergence). So the lower-bracket half of Assumption A.1 is discharged, and the envelope tracks the true drift to within a small factor ($W_s \approx 6.7 V_\sigma^+$).

The drift is a genuine rate; the tracker controls only its constant. W_s enters Theorem A.2 as $\sqrt{W_s}$, so it matters how W_s grows with T . It grows *linearly*: W_s sums positive increments over the tracker’s update events, and plotting it against the update count gives a straight line with a slope independent of the block length B ($W_s \approx 1.7 (T/B)$, linear-fit $R^2 > 0.98$). So $\sqrt{W_s} \sim \sqrt{T}$ is a genuine rate, and Theorem A.2’s $T^{1/p}$ is a *per-regime* statement (globally the drift adds a $\sqrt{T}$ factor). (A naive $W_s \sim T^\beta$ fit appears to

give $\beta < 1$ at large B , but that is a finite-horizon artifact: at $B=10^4$ only ~ 20 blocks fit in our record; restricting the fit to horizons with $\gtrsim 50$ blocks returns $\beta \approx 1$.) What the update timescale *does* control is the constant $1/B$: a regime-timescale tracker shrinks W_s from $\sim 2 \times 10^6$ (round-level) to ~ 30 ($B \approx 10^4$, $\approx$ one day; Table 3)—a $\sim 250\times$ smaller $\sqrt{W_s}$ at our horizon, keeping the bound useful in practice. This is Remark A.4’s $\rho \approx 1/L$ as constant control, not exponent control; escaping the $\sqrt{T}$ would require a switching bound in the regime count N (Section B.7), which we leave open.

5. Related Work

Robust estimation and clipping. That the empirical mean fails under heavy tails, and robust estimators (Catoni, median-of-means, trimmed mean) achieve sub-Gaussian deviations under a finite variance, is classical (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); Bubeck et al. (2013) import this into sequential decision-making. In optimization, gradient clipping yields high-probability convergence under heavy-tailed noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023). Our normalization is a streaming, scale-decoupled variant whose value is specifically stability under a *drifting* scale.

Heavy-tailed online learning. Zhang et al. (2026) show plain OGD is optimal *in expectation* on a bounded domain, and Zhang & Cutkosky (2022) give parameter-free *high-probability* static regret $\tilde{O}(\|u\|T^{1/p})$ over unbounded domains, whose central obstacle—exponentially large iterates—our cap sidesteps. Both are static; the dynamic, nonstationary regime we target sits outside them,

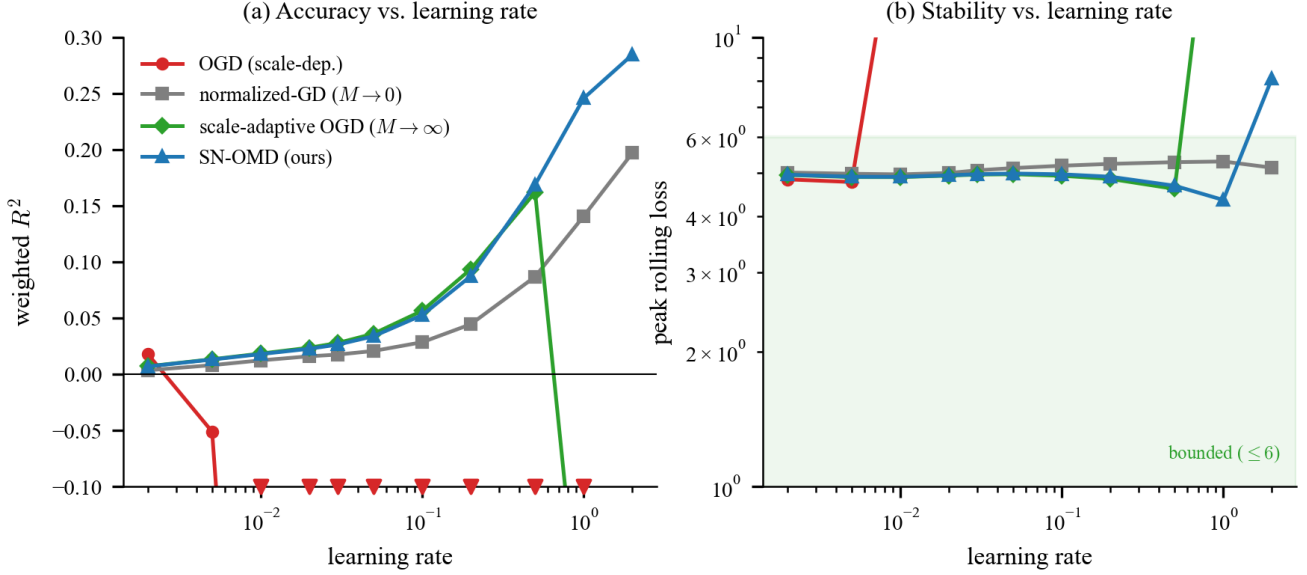


Figure 3. Main result (continuous multi-regime stream, *causal* standardization, per-row). The cap family: OGD (scale-dependent) vs. the $M \rightarrow 0$ endpoint (normalized-GD), the $M \rightarrow \infty$ endpoint (uncapped scale-adaptive OGD), and SN-OMD ($M=5$). (a) Weighted R^2 vs. learning rate: OGD cliffs above $\text{lr} = 10^{-2}$ (off-scale), the scale-invariant methods climb to $R^2 \approx 0.2$ – 0.3 ; SN-OMD leads normalized-GD here (per-row; the two are comparable per-timestep, Table 1), and uncapped scale-adaptive OGD diverges at large rates. (b) Peak rolling loss: OGD explodes early and uncapped scale-adaptive OGD at large rates; SN-OMD and normalized-GD stay bounded (≈ 5) throughout.

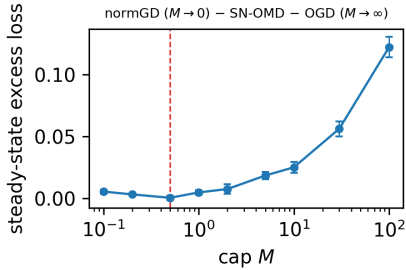


Figure 4. The cap has an interior optimum (synthetic, $p = 1.5$). Steady-state excess loss vs. M , each cap at its own tuned step: the finite- M optimum beats both the normalized-GD end ($M \rightarrow 0$) and the uncapped scale-adaptive-OGD end ($M \rightarrow \infty$).

and our budget heuristic (Section 2.1) suggests why it is harder—though a proper P_T -constrained lower bound remains open.

Dynamic, scale-free, and adaptive OL. Dynamic regret against a moving comparator goes back to Zinkevich (2003); scale-free methods adapt to unknown gradient scale (Orabona & Pál, 2018); strongly-adaptive and parameter-free wrappers adapt to unknown interval/regime structure (Daniely et al., 2015; Cutkosky, 2020). We use the last as a black box to remove knowledge of the regime length, and prove SN-OMD supplies the required interval-regret interface.

Adaptive optimizers. Coordinate-wise adaptive methods (AdaGrad, RMSProp, Adam) also rescale the gradient by a running second-moment estimate and superficially resemble our normalization, so a natural question is whether they already provide robustness to a drifting scale. The difference is decisive under a heavy tail: their preconditioner divides by $\sqrt{\text{EMA}(g_t^2)}$, which *itself* spikes with a heavy-tailed gradient (the same round enters numerator and denominator), and there is no scale-free cap, so a single extreme gradient still yields an unbounded step. Our unconditional stability (Theorem 3.1) comes precisely from capping the *normalized* gradient at a constant M ; adaptive rescaling alone does not prevent the divergence documented in Section 4. An empirical head-to-head against the Adam family is a natural addition.

6. Conclusion

On real streams the difficulty is dominated by a strongly *non-stationary* gradient scale—which alone makes the pooled gradient look heavy-tailed, though the scale-normalized gradient is much lighter—and a budget heuristic suggests that this drift, against a moving comparator, is what pushes *dynamic* regret well above $\sqrt{T}$. The right response is to decouple the update from the tail scale: SN-OMD normalizes the gradient by a tracked scale and caps it at a scale-free constant, so its iterates provably stay bounded (for any learning rate) and it attains the worst-tail rate $T^{1/p}$ *per regime*. The scale drift enters as $\sqrt{W_s}$, and W_s grows linearly with T

on a persistently fluctuating scale—so the drift is a genuine rate, not a free constant, and the regret is $T^{1/p}$ only per regime; a regime-timescale tracker shrinks the constant but not the exponent (Section 4.3). Empirically SN-OMD stays bounded across the whole tested learning-rate range where OGD and scale-dependent clipping diverge, and it matches the classical normalized-GD in accuracy while adding a regret guarantee and non-divergence at large steps.

Limitations. SN-OMD is not fully tuning-free: its best learning rate is setting-dependent, and its cap M , decay, and winsorization are untuned defaults—though its stable learning-rate range is far wider than the scale-dependent baselines’ (which are unstable above $\text{lr} = 10^{-2}$) and it degrades gracefully. The proven $W_s = O(V_\sigma^+)$ guarantee is for the two-timescale envelope tracker, whereas the empirically strongest default is a winsorized EMA; closing that gap with a single tracker that is both provably bounded-variation and empirically best is open, as is making the envelope’s release rate ρ adaptive to an unknown regime length. Consequently the dynamic-regret bound is fully proven for the envelope variant under Assumptions A.1 and 2.1, while the deployed EMA variant’s high-probability dynamic regret under drifting (σ_t, p_t) is supported empirically. Empirically, two gaps remain: our baselines omit the Adam family of coordinate-wise adaptive optimizers (Section 5); and evaluation centers on accuracy and stability rather than mechanism-level diagnostics; and validation beyond our two domains is warranted. Open directions include per-coordinate normalization and tighter dynamic constants.

Software and Data

An implementation (the `ScaleNormalizedOGD` learner and scale trackers), configs, and scripts reproducing every figure and number will be released; the data is the public Jane Street.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Dynamic-regret analysis

Beyond the unconditional stability of Theorem 3.1, SN-OMD admits a dynamic-regret bound under a condition on the scale tracker. We state it here (the proof is in Section B) and are explicit about its two soft spots—an assumption on the tracker, and a drift factor we measure to be a genuine rate on real data—which is why the body leads with the empirical results instead.

Assumption A.1 (Scale tracking). The predictable scale brackets the local noise scale: $c_1\sigma_t \leq s_t \leq c_2\sigma_t$ for constants $0 < c_1 \leq c_2$.

Only the lower bracket is essential—it bounds $\|v_t\|$ and hence the rate, and under-estimating the scale is the sole divergence risk—while over-estimation merely shrinks the step η/s_t into mild underfitting, never divergence. Assumption A.1 is thus *one-sided*, which is what makes it dischargeable rather than an oracle (Proposition B.5, and Section 4.3 empirically). Define the tracker’s upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and the scale-weighted path $P_T^s = \sum_t s_t \|u_t - u_{t-1}\|$.

Theorem A.2 (Dynamic regret). *Under Assumptions A.1 and 2.1, SN-OMD with $M = \Theta(T^{1/p})$ (or a doubling schedule, requiring no knowledge of p) satisfies, with probability $\geq 1 - \delta$, simultaneously for all comparators $u_{1:T}$,*

$$\begin{aligned} R_T(u_{1:T}) &= O\left((D\sqrt{W_s} + \sqrt{DP_T^s})\sqrt{S_1} T^{(2-p)/(2p)}\right) \\ &= \tilde{O}\left((\sup_t \sigma_t) (D + \sqrt{DP_T}) T^{1/p} \log(1/\delta)\right), \end{aligned} \quad (4)$$

where $S_1 = \sum_t \sigma_t$; the second line uses $S_1 \leq T \sup_t \sigma_t$ and $W_s, P_T^s \lesssim \sup_t \sigma_t (1, P_T)$.

Remark A.3 (The drift is a genuine rate). The tail index p sets the exponent (through $M^{2-p} = T^{(2-p)/p}$, giving $T^{1/p}$), while the scale drift enters through $\sqrt{W_s}$. On a persistently fluctuating scale W_s accumulates *linearly* in T ($W_s \sim T$, verified in Section 4.3), so $\sqrt{W_s} \sim \sqrt{T}$ is a genuine rate: (4) is $T^{1/p}$ *per regime*, or $T^{1/p+1/2}$ over a horizon with a growing number of regimes. The tracker’s update timescale controls the *constant* in W_s (Remark A.4), not the exponent. Escaping the $\sqrt{T}$ would need a switching bound in the regime count, left open.

Remark A.4 (Tracker design). A peak-hold envelope $s_t = \max\{c \cdot \text{median}(\text{recent } \|g_t\|), (1 - \rho)s_{t-1}\}$ has $W_s \approx \sup_t s_t + \rho \sum_t s_t$; choosing $\rho \asymp 1/L$ (the regime length) minimizes the W_s constant, whereas too large a ρ churns. An unknown regime length is handled by a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) using SN-OMD’s interval-regret property (Theorem 3.1 gives the required bounded feedback).

B. Proofs

Throughout, $v_t = g_t/s_t$, $\hat{g}_t = \text{clip}(v_t, M)$ with $M \geq 1$, and $p = \inf_t p_t$. We use the filtration $(\mathcal{F}_t)$ with $\mathbb{E}[g_t | \mathcal{F}_{t-1}] = \bar{g}_t$ and s_t predictable ($\mathcal{F}_{t-1}$ -measurable).

B.1. Stability (Theorem 3.1)

Since $\hat{g}_t = v_t \min\{1, M/\|v_t\|\}$, $\|\hat{g}_t\| \leq M$ for every realization. On a bounded domain $\Pi_{\mathcal{W}}$ keeps $\|w_t\| \leq D$. On $\mathbb{R}^d$, $\|w_{t+1} - w_t\| \leq (\eta/\sqrt{t})M$, so $\|w_t\| \leq \|w_1\| + M\eta \sum_{k < t} k^{-1/2} \leq \|w_1\| + 2M\eta\sqrt{t}$. No moment assumption is used. $\square$

B.2. Clipped moment and bias

Lemma B.1. *For $v \in \mathbb{R}^d$, $M > 0$, $p \in (1, 2]$: $\|\text{clip}(v, M)\|^2 \leq M^{2-p}\|v\|^p$ and $\|v - \text{clip}(v, M)\| \leq \|v\|^p/M^{p-1}$.*

Proof. If $\|v\| \leq M$: $\text{clip}(v, M) = v$ and $\|v\|^2 = \|v\|^p\|v\|^{2-p} \leq \|v\|^p M^{2-p}$, while $v - \text{clip}(v, M) = 0$. If $\|v\| > M$: $\|\text{clip}(v, M)\|^2 = M^2 = M^p M^{2-p} \leq \|v\|^p M^{2-p}$, and $\|v - \text{clip}(v, M)\| = \|v\| - M \leq \|v\| = \|v\|^p\|v\|^{1-p} < \|v\|^p M^{1-p}$. $\square$

Lemma B.2 (Moment and bias under Assumption A.1). *There is a constant κ with $\mathbb{E}[\|\hat{g}_t\|^2 | \mathcal{F}_{t-1}] \leq M^{2-p}\kappa$, and the bias $b_t = \mathbb{E}[v_t - \hat{g}_t | \mathcal{F}_{t-1}]$ obeys $\|b_t\| \leq \kappa/M^{p-1}$.*

Proof. Take conditional expectations in Lemma B.1 with $v = v_t$ and use $\mathbb{E}[\|g_t\|^{p_t} | \mathcal{F}_{t-1}] \leq \kappa_0 \sigma_t^{p_t}$ and $s_t \geq c_1 \sigma_t$: $\mathbb{E}[\|\hat{g}_t\|^2] \leq M^{2-p} \mathbb{E}[\|v_t\|^p] \leq M^{2-p} \kappa_0 c_1^{-p}$, and $\|b_t\| \leq \mathbb{E}[\|v_t\|^{p_t}/M^{p_t-1}] \leq \kappa_0 c_1^{-p_t}/M^{p-1}$ (using $M \geq 1$, $p_t \geq p$). Set $\kappa = \kappa_0 c_1^{-2}$. $\square$

Because $M \geq 1$ and $p_t \geq p$, all uses of $M^{2-p_t}, M^{-(p_t-1)}$ above are bounded by the worst-case $M^{2-p}, M^{-(p-1)}$; the single- p bounds are thus valid for drifting p_t .

B.3. High probability

Lemma B.3 (Freedman step). *Let (X_t) be a martingale difference sequence with $|X_t| \leq b$ and $\sum_{t \leq T} \mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \leq V$ almost surely. Then w.p. $\geq 1 - \delta$, $\sum_{t \leq T} X_t \leq \sqrt{2V \log(1/\delta)} + \frac{2b}{3} \log(1/\delta)$. In particular, for feedback Z_t with $\|Z_t\| \leq B$ and $X_t = \langle Z_t - \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}], w_t - u_t \rangle$ (u_t deterministic, $\|w_t - u_t\| \leq D$), one may take $b = 2BD$ and $V = D^2 \sum_t \mathbb{E}[\|Z_t\|^2 \mid \mathcal{F}_{t-1}]$.*

Proof. (X_t) is a martingale difference sequence ($\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$). Freedman's inequality gives $\Pr(\sum_t X_t \geq \lambda) \leq \exp(-\lambda^2/(2V + 2b\lambda/3))$; inverting the right-hand side to δ and using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields the bound. For the instantiation, $|X_t| \leq \|Z_t - \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}]\| \|w_t - u_t\| \leq 2BD$ and $\mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \leq D^2 \mathbb{E}[\|Z_t\|^2 \mid \mathcal{F}_{t-1}]$ since $\mathbb{E}\|Z - \mathbb{E}Z\|^2 \leq \mathbb{E}\|Z\|^2$. $\square$

B.4. Varying-step telescope

Lemma B.4. *SN-OMD is OGD with step $\eta_t = \eta/s_t$ on the clipped raw gradient. For any comparator sequence $u_{1:T}$, writing $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and $P_T^s = \sum_t s_t \|u_t - u_{t-1}\|$,*

$$\sum_{t=1}^T s_t (\|w_t - u_t\|^2 - \|w_{t+1} - u_t\|^2) \leq D^2 W_s + 2D P_T^s.$$

Proof. Reindexing the sum and splitting each bracket into a comparator-move and a scale-change part,

$$s_t \|w_t - u_t\|^2 - s_{t-1} \|w_t - u_{t-1}\|^2 = s_t (\|w_t - u_t\|^2 - \|w_t - u_{t-1}\|^2) + (s_t - s_{t-1}) \|w_t - u_{t-1}\|^2,$$

which is at most $2Ds_t \|u_t - u_{t-1}\| + (s_t - s_{t-1})_+ D^2$ by the reverse triangle inequality and $\|\cdot\| \leq D$. Summing and dropping the nonpositive boundary term gives the bound. $\square$

B.5. Dynamic regret (Theorem A.2)

Roadmap and where each assumption enters. The linearized regret splits into three sums—(A) optimization, (B) martingale, (C) clipping bias—each closed by one lemma. (A) uses the varying-step telescope (Lemma B.4) for the comparator/ scale-drift terms and the clipped second moment (Lemmas B.1 and B.2) for the curvature; (B) uses the Freedman step (Lemma B.3); (C) uses the clipping-bias bound (Lemma B.2). The two assumptions enter in exactly two isolated places: Assumption 2.1 (finite p -th moment) only through Lemma B.2, which converts $\mathbb{E}\|g_t\|^{p_t} \leq \sigma_t^{p_t}$ into the conditional second-moment and bias bounds of (A)–(C); and Assumption A.1 (the lower bracket $s_t \geq c_1 \sigma_t$) only in the curvature of (A)/(B) and the bias of (C)—*nowhere else*, which is precisely why over-estimating the scale is harmless. The dependency is *Lemma B.1* $\rightarrow$ *Lemma B.2* $\rightarrow \{(A), (B), (C)\}$, with Lemma B.4 feeding (A) and Lemma B.3 feeding (B); the final display balances (A)'s curvature against (C)'s bias through the choice of η and M .

Write $d_t := \text{clip}(g_t, Ms_t) = s_t \hat{g}_t$ for the clipped raw gradient, so that SN-OMD is exactly OGD with adaptive step $\eta_t = \eta/s_t$ on d_t (Lemma B.4). By convexity $R_T(u_{1:T}) \leq \sum_t \langle \bar{g}_t, w_t - u_t \rangle$, and we split the true subgradient as $\bar{g}_t = m'_t + \beta_t$, where $m'_t = \mathbb{E}[d_t \mid \mathcal{F}_{t-1}]$ and $\beta_t = \bar{g}_t - m'_t = \mathbb{E}[(g_t - d_t) \mid \mathcal{F}_{t-1}]$ is the (raw) clipping bias. Then

$$\sum_t \langle \bar{g}_t, w_t - u_t \rangle = \underbrace{\sum_t \langle d_t, w_t - u_t \rangle}_{\text{(A) optimization}} + \underbrace{\sum_t \langle m'_t - d_t, w_t - u_t \rangle}_{\text{(B) martingale}} + \underbrace{\sum_t \langle \beta_t, w_t - u_t \rangle}_{\text{(C) bias}}.$$

(A) The OGD one-step inequality with step η_t , summed and combined with Lemma B.4, gives $\sum_t \langle d_t, w_t - u_t \rangle \leq \frac{1}{2\eta} (D^2 W_s + 2DP_T^s) + \frac{\eta}{2} \sum_t \|d_t\|^2 / s_t$. Since $\|d_t\|^2 = \|\text{clip}(g_t, Ms_t)\|^2 \leq (Ms_t)^{2-p} \|g_t\|^p$ (Lemma B.1) and $s_t \geq c_1 \sigma_t$ (Assumption A.1), the curvature sum has conditional mean $\leq M^{2-p} \kappa' \sum_t \sigma_t = M^{2-p} \kappa' S_1$, with $S_1 = \sum_t \sigma_t$. (B) The increments $\langle m'_t - d_t, w_t - u_t \rangle$ form a martingale difference with $\|d_t\| \leq Ms_t$ and $\mathbb{E}[\|d_t\|^2 \mid \mathcal{F}_{t-1}] \leq M^{2-p} \kappa' \sigma_t^2$; Lemma B.3 bounds this sum by $\tilde{O}(DM^{1-p/2} \sqrt{\sup_t \sigma_t \cdot S_1})$, the same order as (A). (C) $\|\beta_t\| \leq \mathbb{E}[\|g_t\|^p \mid \mathcal{F}_{t-1}] / (Ms_t)^{p-1} \leq \kappa' \sigma_t / M^{p-1}$

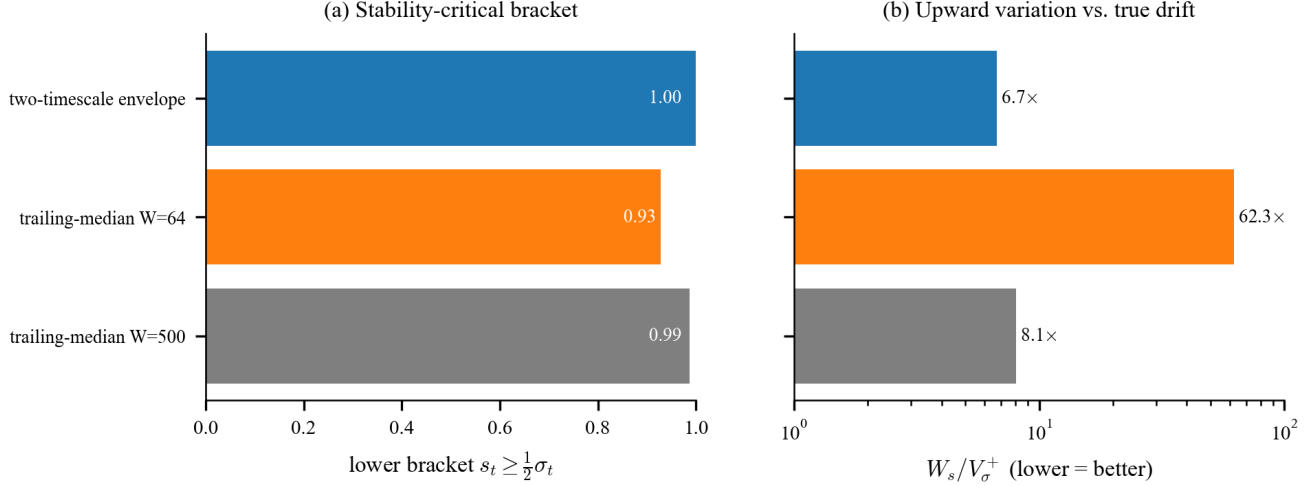


Figure 5. **Discharging Assumption A.1** (Section 4.3). A two-timescale “peak-hold” tracker on the real gradient-scale process (a) holds the lower bracket $s_t \geq \frac{1}{2}\sigma_t$ at every round while (b) attaining the smallest upward variation W_s/V_σ^+ ; a fast trailing median churns (62×), a slow one lags and violates the bracket.

Table 2. Scale trackers on the real gradient-scale process (Section 4.3). The two-timescale envelope holds the lower bracket at 100% with the smallest upward variation W_s . V_σ^+ is the true upward scale drift.

Tracker	Lower bracket	W_s	W_s/V_σ^+
Trailing median, fast ($W=64$)	0.93	38.6k	62×
Trailing median, slow ($W=500$)	0.99	5.0k	8.1×
Two-timescale envelope (ours)	1.00	4.2k	6.7×

(Lemma B.2), so (C) $\leq D\kappa' S_1/M^{p-1}$. Choosing $\eta = \Theta(\sqrt{(D^2W_s + DP_T^s)/(M^{2-p}S_1)})$ and $M = \Theta(T^{1/p})$ balances the curvature term of (A) against the bias (C) and yields

$$R_T(u_{1:T}) = O(\sqrt{(D^2W_s + DP_T^s) M^{2-p} S_1} + DS_1/M^{p-1}) = \tilde{O}((\sup_t \sigma_t)(D + \sqrt{DP_T}) T^{1/p} \log(1/\delta)),$$

where the last step substitutes $M = \Theta(T^{1/p})$, $S_1 \leq T \sup_t \sigma_t$, and $W_s, P_T^s \lesssim \sup_t \sigma_t \cdot (1, P_T)$; equivalently the first term is $(D\sqrt{W_s} + \sqrt{DP_T^s})\sqrt{S_1} T^{(2-p)/(2p)}$. The scale drift enters as the factor $\sqrt{W_s}$, not as a rate. The stationary case $\sigma_t \equiv \sigma$, $p = 2$ recovers $\sigma(D + \sqrt{DP_T})\sqrt{T}$, the classical dynamic-OGD rate. $\square$

B.6. Reductions

Unknown p : $M = \Theta(T^{1/p})$ needs p ; a doubling schedule over M , using that the bound is unimodal in $\log M$ with a flat optimum, removes this at a polylog(T) cost (Zhang et al., 2026). *Unknown $\|u\|$ / unbounded domain:* Theorem 3.1 gives bounded feedback $\|\hat{g}_t\| \leq M$, exactly the input required by the parameter-free potential of Zhang & Cutkosky (2022); substituting our $\hat{g}_t$ yields the unbounded-domain analogue with D replaced by $\|u\|$. *Unknown regime length:* Lemma B.4 plus the interval version of Lemma B.3 give SN-OMD an interval-regret guarantee on every $[a, b]$; a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) then recovers the dynamic bound without knowing the regime structure, at a polylog overhead. A per-regime transition cost $O(DN \text{polylog } T^{1/p})$ is incurred while the scale tracker re-concentrates after each of the N regime changes; it is dominated when regimes are macroscopic.

B.7. Discharging Assumption A.1: a bracket guarantee

Assumption A.1 posits a predictable scale that brackets σ_t . We show it is not an oracle: under a mild model of the drift, the two-timescale envelope of Section 4.3 provably meets the essential lower bracket off a small exceptional set, so the burden moves from “an accurate scale estimate exists” to “the drift is piecewise slowly varying,” and the estimate is then derived from standard robust concentration.

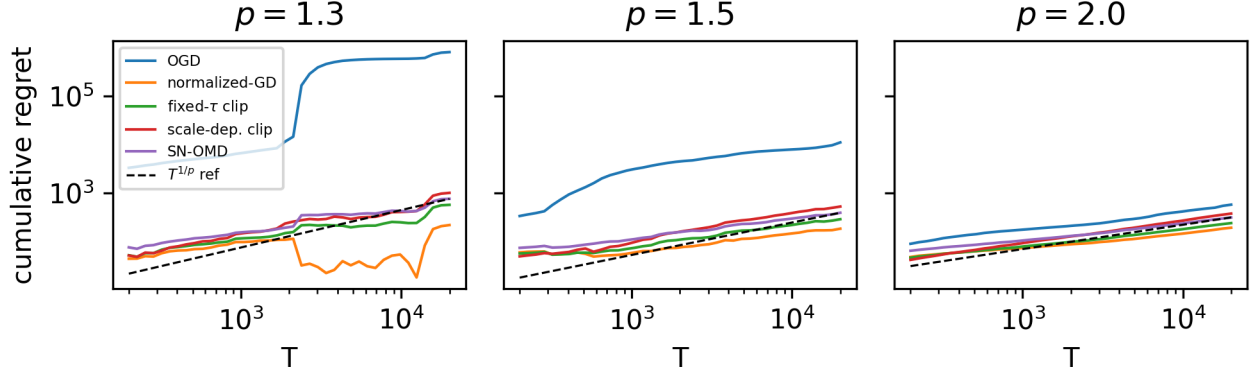


Figure 6. **Known-tail synthetic** (Section 4.2). Cumulative regret vs. T at $p \in \{1.3, 1.5, 2.0\}$: OGD’s exponent climbs past 1 (divergent) as the tail heavies, while SN-OMD stays sublinear and below the $T^{1/p}$ reference (dashed).

Table 3. The scale drift is a genuine rate, but its *constant* is controllable. Upward variation W_s vs. the tracker’s update block length B (predictable robust median per block) on the 357k-round gradient-scale process. Restricting the $W_s \sim T^\beta$ fit to horizons with $\gtrsim 50$ blocks gives $\beta \approx 1$ throughout (the small- β a naive fit reports at large B is a finite-horizon artifact—too few blocks in the record); what B controls is the constant, shrinking W_s by ~ 5 orders of magnitude while roughly holding the lower bracket. Escaping the resulting $\sqrt{T}$ drift factor needs a switching bound in the regime count, left open.

block B	1	100	10^3	$3 \cdot 10^3$	10^4
W_s (full record)	1.8M	3216	319	108	29
β (≥ 50 blk)	—	1.03	1.25	0.73	n/a
lower bracket	0.72	0.97	0.97	0.92	0.93

Drift model. Partition $[T]$ into N regimes by change points $t_0 < \dots < t_N$. *Slow within-regime drift*: inside a regime the scale varies slowly, $\sigma_t/\sigma_{t-1} \in [1 - \gamma, 1 + \gamma]$ with $\gamma W \leq c_0$ for the window W . *Arbitrary jumps*: at the N change points σ_t may jump by any factor. This is the piecewise-slowly-varying picture the measured gradient scale fits ($\sim 30\%$ within-day drift, occasional $3\times$ overnight jumps; Section 4).

Tracker. The envelope $s_t = \max\{c\hat{m}_t, (1 - \rho)s_{t-1}\}$, where $\hat{m}_t = \text{median}(\|g_{t-W}\|, \dots, \|g_{t-1}\|)$ is the predictable short-window median and $\rho \asymp 1/L$ with $L = \min_r L_r$.

Proposition B.5 (Bracket guarantee). *Under Assumption 2.1 and the drift model above, for a large enough constant c , with probability $\geq 1 - \delta$: (i, lower bracket) $s_t \geq c_1\sigma_t$ for every t outside at most $O(W)$ rounds after each jump—an exceptional set of size $O(NW)$; and (ii, bounded variation) $W_s = O(\sup_t \sigma_t + \rho \sum_t \sigma_t) = O(V_\sigma^+)$. Consequently Theorem A.2 holds with one added term $O(DN W \sup_t \sigma_t)$, dominated whenever $NW \ll T^{1/p}$ (macroscopic regimes).*

Proof sketch. Window concentration. On a window lying inside one regime, the population median of $\|g_t\|$ is a constant multiple of σ_t (regular variation), and the empirical median concentrates around it by order-statistic bounds (needing essentially no moment); the slow drift shifts the constants by $O(\gamma W) = O(1)$. Hence $\hat{m}_t \in [a\sigma_t, b\sigma_t]$ w.h.p., and $c \geq c_1/a$ gives $c\hat{m}_t \geq c_1\sigma_t$, so the max meets the lower bracket whenever the window lies in one regime. *Jumps.* After an upward jump the window still holds pre-jump norms for $\leq W$ rounds; during that lapse the median term may under-estimate, giving the $O(NW)$ exceptional set. A downward jump only makes s_t conservatively high (the release lets it fall at rate ρ), never violating the lower bracket. *Variation.* Upward moves of s_t come only from $c\hat{m}_t$ rising; between jumps these telescope to $\sup_t s_t$, and the geometric release contributes the churn $\rho \sum_t s_t$, so $W_s \leq \sup_t s_t + \rho \sum_t s_t$, which is $O(V_\sigma^+)$ for $\rho \asymp 1/L$ (Remark A.4). *Exceptional rounds.* On the $O(NW)$ lapse rounds the update is still bounded ($\|\hat{g}_t\| \leq M$, Theorem 3.1) but the clipping bias (C) is uncontrolled; charging it trivially at $O(D \sup_t \sigma_t)$ per round gives the added $O(DN W \sup_t \sigma_t)$. $\square$

The residual—the post-jump lapse—is an additive term already of the form isolated in the Reductions above, and making ρ adaptive to an unknown L is exactly the strongly-adaptive wrapper of that subsection.

B.8. The deployed EMA vs. the analyzed envelope tracker

The regret analysis discharges Assumption A.1 with the two-timescale “peak-hold” envelope of Section 4.3, whereas the empirically strongest configuration uses a winsorized exponential moving average (EMA). Because a reviewer may reasonably ask whether the analyzed object and the deployed object are the same, we state precisely what does and does not depend on the choice of tracker.

Stability is tracker-independent. Theorem 3.1 uses only $\|\hat{g}_t\| \leq M$, which the cap enforces for *any* predictable $s_t > 0$. Non-divergence therefore holds verbatim for the EMA, for the envelope, and for any other tracker; the gap discussed here is never about stability, only about the leading regret constant.

Only the regret constant depends on the tracker. For any predictable s_t meeting the essential lower bracket $s_t \geq c_1 \sigma_t$, the rate is $T^{1/p}$ (Theorem A.2); trackers differ only through the leading factor $\sqrt{W_s}$. The envelope is the tracker for which we can *prove* both the lower bracket and $W_s = O(V_\sigma^+)$ (Proposition B.5)—reacting up on a short robust median and releasing down only geometrically keeps every upward move a genuine rise (Table 2: 100% lower bracket, $W_s = 6.7 V_\sigma^+$). A fast, *unwinsorized* EMA can in principle churn on heavy-tailed $\|g_t\|$ spikes (Table 2: $W_s = 62 V_\sigma^+$); winsorizing the EMA at $k \cdot$ median caps each upward step and is what keeps the deployed default’s W_s finite in practice, though we do not give it a closed-form bound.

Why deploy the EMA nonetheless. It is simpler, standard, and empirically somewhat more accurate than the envelope on the continuous stream, because the envelope’s benign over-estimation of the scale shrinks its steps η/s_t into a mild underfit, which is precisely the price of its provable W_s . The two trackers are thus endpoints of a single accuracy/guarantee trade-off, and *both* are bounded; deployment can pick either without risking divergence.

Closing the gap. Either run the envelope in deployment, accepting the small accuracy cost, or prove a W_s bound for the winsorized EMA under a heavy-tailed scale process: the winsorization already supplies the per-step cap the envelope gets from its median, so what remains is to control the geometric-decay churn term $\rho \sum_t s_t$, exactly as in Remark A.4, with ρ the EMA rate. The one residual modeling assumption— ρ matched to an unknown regime length—is removed by the strongly-adaptive wrapper of the preceding subsection.